Practical No. 06

Q.1] Write and Execute code to display selected hobbies using CheckBox.

Activity_main.xml:-

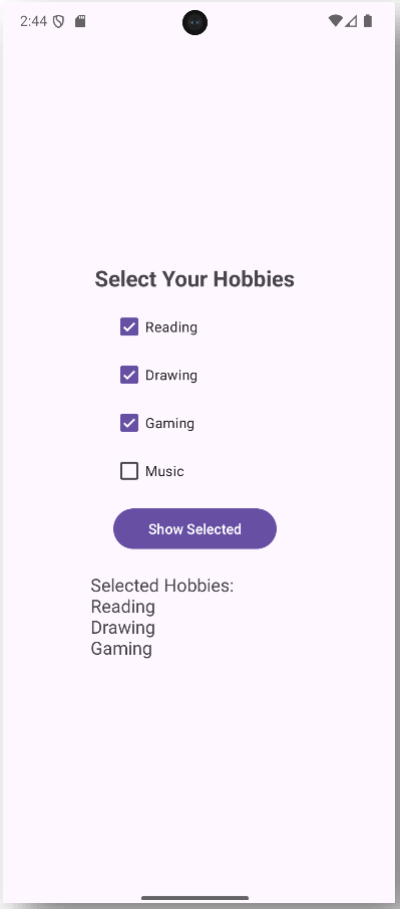
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:id="@+id/main"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:padding="20dp">

   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Select Your Hobbies"
      android:textSize="22sp"
      android:textStyle="bold"
      android:layout_marginBottom="10dp"/>

   <CheckBox
      android:id="@+id/cbReading"
      android:text="Reading"
      android:ems="10"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"/>

   <CheckBox
      android:id="@+id/cbDrawing"
      android:text="Drawing"
      android:ems="10"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"/>

   <CheckBox
      android:id="@+id/cbGaming"
```

```
        android:text="Gaming"
        android:ems="10"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

    <CheckBox
        android:id="@+id/cbMusic"
        android:text="Music"
        android:ems="10"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/btnShow"
        android:text="Show Selected"
        android:ems="10"
        android:layout_margin="10dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

    <TextView
        android:id="@+id/txtResult"
        android:ems="10"
        android:textSize="18sp"
        android:text="Selected Hobbies"
        android:layout_margin="10dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>
</LinearLayout>
```

MainActivity.java:

```
package com.techtitans.madpracticals;

import android.annotation.SuppressLint;
import android.os.Bundle;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.TextView;
```

```
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        CheckBox cbReading = findViewById(R.id.cbReading);
        CheckBox cbDrawing = findViewById(R.id.cbDrawing);
        CheckBox cbGaming  = findViewById(R.id.cbGaming);
        CheckBox cbMusic   = findViewById(R.id.cbMusic);

        Button btnShow = findViewById(R.id.btnShow);
        TextView txtResult = findViewById(R.id.txtResult);

        btnShow.setOnClickListener(v -> {
            String result = "Selected Hobbies:";

            if (cbReading.isChecked())
                result += "\nReading";

            if (cbDrawing.isChecked())
                result += "\nDrawing";

            if (cbGaming.isChecked())
                result += "\nGaming";

            if (cbMusic.isChecked())
                result += "\nMusic";

            txtResult.setText(result);
        });
    }
}
```



Q.2] Write and Execute code to count how many CheckBoxes are selected and display the count.

Activity_main.xml:-

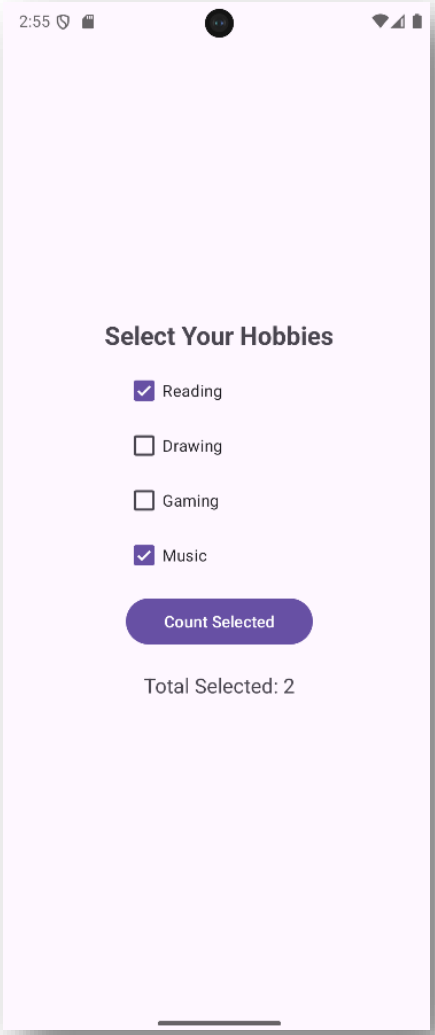
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:id="@+id/main"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:padding="20dp">

   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Select Your Hobbies"
      android:textSize="22sp"
      android:textStyle="bold"
      android:layout_marginBottom="10dp"/>

   <CheckBox
      android:id="@+id/cbReading"
      android:text="Reading"
      android:ems="10"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"/>

   <CheckBox
      android:id="@+id/cbDrawing"
      android:text="Drawing"
      android:ems="10"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"/>

   <CheckBox
      android:id="@+id/cbGaming"
      android:text="Gaming"
      android:ems="10"
```

```
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

    <CheckBox
        android:id="@+id/cbMusic"
        android:text="Music"
        android:ems="10"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/btnCount"
        android:text="Count Selected"
        android:ems="10"
        android:layout_margin="10dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

    <TextView
        android:id="@+id/txtCount"
        android:text="Total Selected Hobbies : 0"
        android:textSize="18sp"
        android:layout_margin="10dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>
</LinearLayout>
```

MainActivity.java:

```
package com.techtitans.madpracticals;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.TextView;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
```

```
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        CheckBox cbReading = findViewById(R.id.cbReading);
        CheckBox cbDrawing = findViewById(R.id.cbDrawing);
        CheckBox cbGaming  = findViewById(R.id.cbGaming);
        CheckBox cbMusic   = findViewById(R.id.cbMusic);

        Button btnCount = findViewById(R.id.btnCount);
        TextView txtCount = findViewById(R.id.txtCount);

        btnCount.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {

                    int count = 0;
                if (cbReading.isChecked()) {
                    count++;
                }
                if (cbDrawing.isChecked()) {
                    count++;
                }
                if (cbGaming.isChecked()) {
                    count++;
                }
                if (cbMusic.isChecked()) {
                    count++;
                }
                txtCount.setText("Total Selected: " + count);
            }
        });
    }
}
```



Q.3] Write and Execute code to clear all CheckBox and RadioButton selections on button click.

Activity_main.xml:-

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:id="@+id/main"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:gravity="center"
   android:orientation="vertical"
   android:padding="20dp">

   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Select Hobbies"
      android:layout_marginStart="-40dp"
      android:textStyle="bold"
      android:textSize="20sp"
      android:layout_marginBottom="8dp"/>

   <CheckBox
      android:id="@+id/cbReading"
      android:text="Reading"
      android:ems="10"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"/>

   <CheckBox
      android:id="@+id/cbDrawing"
      android:text="Drawing"
      android:ems="10"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"/>

   <CheckBox
      android:id="@+id/cbGaming"
      android:text="Gaming"
```

```
    android:ems="10"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"/>

  <CheckBox
    android:id="@+id/cbMusic"
    android:text="Music"
    android:ems="10"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"/>

  <TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Select Gender"
    android:textStyle="bold"
    android:textSize="20sp"
    android:layout_marginStart="-40dp"
    android:layout_marginTop="12dp"
    android:layout_marginBottom="6dp"/>

  <RadioGroup
    android:id="@+id/rgGender"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content">

    <RadioButton
      android:id="@+id/rbMale"
      android:text="Male"
      android:ems="10"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"/>

    <RadioButton
      android:id="@+id/rbFemale"
      android:text="Female"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"/>

    <RadioButton
      android:id="@+id/rbOther"
```

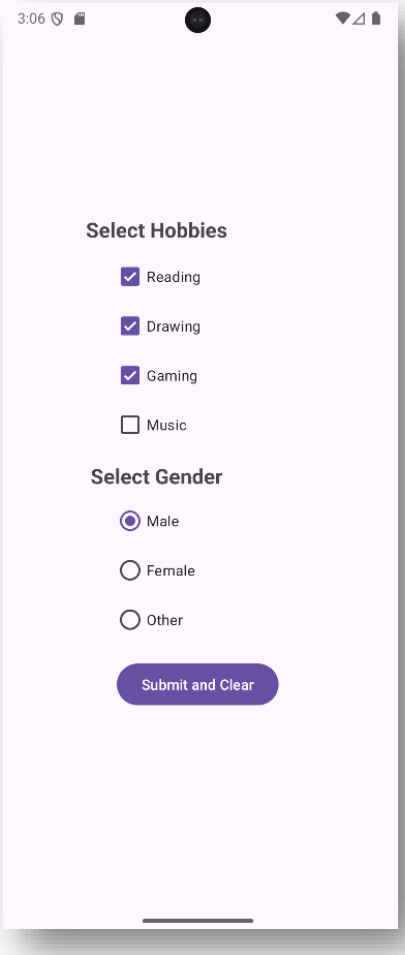
```
        android:text="Other"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>
    </RadioGroup>

    <Button
        android:id="@+id/btnClear"
        android:text="Submit and Clear"
        android:layout_marginTop="15dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>
</LinearLayout>
```

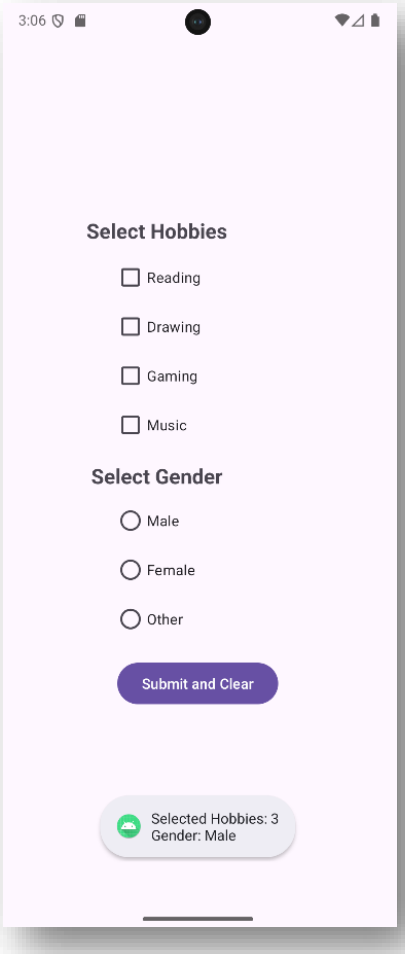
MainActivity.java:

```
package com.techtitans.madpracticals;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.RadioButton;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        CheckBox cbReading = findViewById(R.id.cbReading);
        CheckBox cbDrawing = findViewById(R.id.cbDrawing);
        CheckBox cbGaming = findViewById(R.id.cbGaming);
```

```
CheckBox cbMusic = findViewById(R.id.cbMusic);

RadioButton rbMale = findViewById(R.id.rbMale);
RadioButton rbFemale = findViewById(R.id.rbFemale);
RadioButton rbOther = findViewById(R.id.rbOther);

Button btnClear = findViewById(R.id.btnClear);

// FIX 2: Cleaned up the braces for the OnClickListener
btnClear.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        int count = 0;
        String gender = "Not Selected";

        if (cbReading.isChecked()) {
            count++;
            cbReading.setChecked(false);
        }
        if (cbDrawing.isChecked()) {
            count++;
            cbDrawing.setChecked(false);
        }
        if (cbGaming.isChecked()) {
            count++;
            cbGaming.setChecked(false);
        }
        if (cbMusic.isChecked()) {
            count++;
            cbMusic.setChecked(false);
        }
        if (rbMale.isChecked()) {
            gender = "Male";
            rbMale.setChecked(false);
        }
        if (rbFemale.isChecked()) {
            gender = "Female";
            rbFemale.setChecked(false);
        }
        if (rbOther.isChecked()) {
            gender = "Other";
```

```
                rbOther.setChecked(false);
            }
            Toast.makeText(MainActivity.this,"Selected Hobbies: " + count + "\nGender: " +
gender, Toast.LENGTH_LONG).show();
        }
    });
  }
}
```





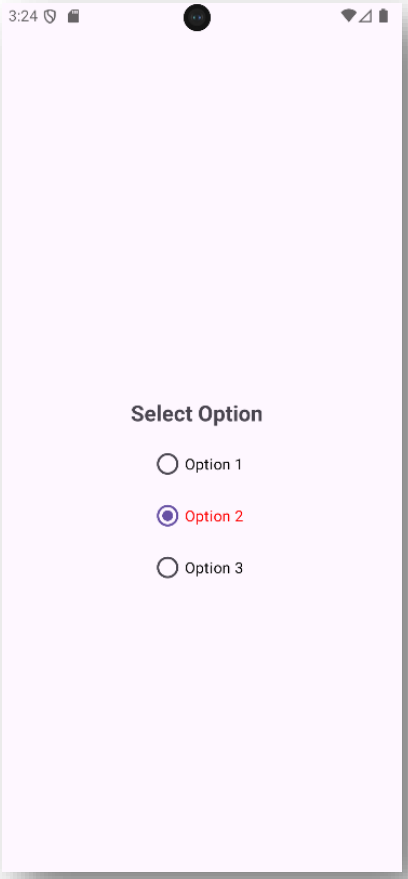
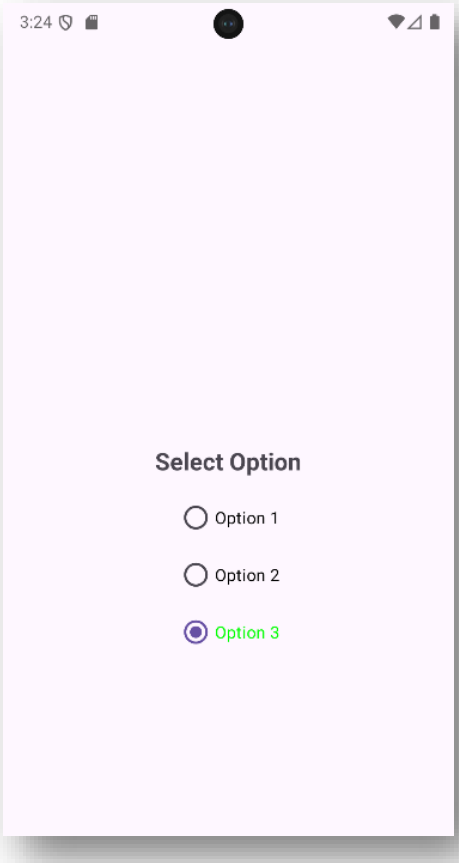
Q.4] Write code to change the text color of a RadioButton when selected.

Activity_main.xml:-

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
  xmlns:android="http://schemas.android.com/apk/res/android"
  android:id="@+id/main"
```

```
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Select Option"
        android:textSize="20sp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"/>

    <RadioGroup
        android:id="@+id/rgOptions"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content">

        <RadioButton
            android:id="@+id/rbOption1"
            android:text="Option 1"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"/>

        <RadioButton
            android:id="@+id/rbOption2"
            android:text="Option 2"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"/>

        <RadioButton
            android:id="@+id/rbOption3"
            android:text="Option 3"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"/>
    </RadioGroup>
</LinearLayout>
```

MainActivity.java:

```
package com.techtitans.madpracticals;

import android.graphics.Color;
import android.os.Bundle;
import android.widget.RadioButton;
import android.widget.RadioGroup;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main),
                (v, insets) -> {
                    Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
                    v.setPadding(systemBars.left, systemBars.top,
                            systemBars.right, systemBars.bottom);
                    return insets;
                });

        RadioGroup rgOptions = findViewById(R.id.rgOptions);

        RadioButton rb1 = findViewById(R.id.rbOption1);
        RadioButton rb2 = findViewById(R.id.rbOption2);
        RadioButton rb3 = findViewById(R.id.rbOption3);

        rgOptions.setOnCheckedChangeListener(new RadioGroup.OnCheckedChangeListener() {
            @Override
            public void onCheckedChanged(RadioGroup group, int checkedId) {
```

```
            // reset all colors
            rb1.setTextColor(Color.BLACK);
            rb2.setTextColor(Color.BLACK);
            rb3.setTextColor(Color.BLACK);

            if (checkedId == R.id.rbOption1) {
                rb1.setTextColor(Color.BLUE);
            }

            if (checkedId == R.id.rbOption2) {
                rb2.setTextColor(Color.RED);
            }

            if (checkedId == R.id.rbOption3) {
                rb3.setTextColor(Color.GREEN);
            }
        }
    });
    }
}
```